DEPARTMENT OF COMPUTER SCIENCE

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PROJECT PROPOSAL

INNsight

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Course Title:	Software Design and Analysis
Semester:	4th - Spring
Date of Submission:	8th February 2026
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1. Project Overview

INNsight is a high-tech, AI-driven hotel booking software designed to change the customer experience of hospitality services. INNsight is a virtual concierge that interacts with customers instead of using the traditional forms of static booking. The system combines a robust Java-based Graphical User Interface (GUI) and Generative AI to create a system capable of natural and conversational interactions that simulate human interactions. With the ability to discern the desires of the customers and process real-time data, INNsight enables the simplification of the entire reservation process, beginning with the initial request and concluding with the receipt of a booking confirmation, which offers consumers a unified and thoroughly independent experience.

2. Motivation

The hospitality industry is entering a stage where speed and accuracy is paramount. This notwithstanding, the conventional hotel management systems are still struggling with a number of challenges that will be overcome by INNsight. Limited availability and slow response times present one of the biggest problems since the human staff is not able to work full time. In due course, the customers who make inquiries at late hours or during peak time often undergo long waiting time, and this could lead to loss of business to the hotel. Human error is another problem and this arises with books being handled manually. The receptionists can misspell customer names, insert a wrong date or accidentally book two guests in one room thereby causing dissatisfaction. Also, existing systems are not automated, thus, increasing the pace of operations and leaning more towards manual operations. With ever-changing technology, there is an increased demand of leaner systems capable of automatically executing routine assignments so that businesses may run effectively without necessarily being under human care. The solutions to these issues as presented by INNsight will be to utilize artificial intelligence to offer real-time answers around the clock, automation removes typing errors and minimizes time wasted on booking to a few seconds instead of a few minutes.

3. Objectives

The major goal of this project is to create INNsight as a desktop application which is capable of allowing the customers to book rooms at the hotel by conversing with the application using natural language. The other objective is to adopt a relational database back-end, e.g. , MySQL or SQLite to guarantee the safe and permanent storage of room inventory records and client information. The

project will also endeavor to develop a user-friendly Java Swing or JavaFX graphical interface which will be used to bind the conversational interface where users lacking technical knowledge can easily interface with high-tech technology. Additionally, the project will show how the artificial intelligence can automate business processes based on the state of the system, which will guarantee the adequate checking of the rooms availability in real-time and the absence of the possibility to make two reservations.

4. Expected Outcomes

The system is supposed to deliver real-time responses to the customers any time and no question is left unanswered. Automation will eliminate human mistakes in the system that will ensure that customer names, their booking dates, and room details are properly saved in the database at all times. The efficiency of operational processes will also increase because the reservation process will be made fast and easy without involving the staff until the guest arrives. Also, the availability of rooms will be updated on the database every time a reservation is made in real time. The project will also be used as an example showing how hotels can shift their manual operations to complete automation and operational systems based on artificial intelligence.

5. Project Scope

As the automation and artificial intelligence continue to enter the different sectors, so too is the hospitality sector. The network of INNsight is designed to facilitate and spearhead this change by offering fully independent system of reservations. The project is made up of a number of necessary modules which collaborate to provide this objective. It has a smart chat interface that provides a user-friendly GUI with which a user can communicate using natural language. It also has an AI logic core that interprets user input and retrieves the required booking information about reservation dates, room preferences, and contact information. The system also includes database management based on the well-organized SQL schema that processes the data on the rooms, customers, and reservations in progress. It also has a business logic layer which is a Java-based algorithm that is used to handle conflicts in schedules, pricing and data validation.

6. Tools and Technologies

The system will be graphical user interface (Java Swing or JavaFX) based and the back end database functionality will be Java SQL or SQLite.